In [1]:
```
!pip install ultralytics opencv-python
```

```
Collecting ultralytics
  Downloading ultralytics-8.4.138-py3-none-any.whl.metadata (45 kB)
Collecting opencv-python
  Using cached opencv_python-5.0.0.93-cp37-abi3-win_amd64.whl.metadata (20 kB)
Requirement already satisfied: filelock>=3.16.1 in E:\OPG_env\opg_venv\Lib\site-packages (from ultralytics) (3.20.0)
Requirement already satisfied: numpy>=1.23.0 in E:\OPG_env\opg_venv\Lib\site-packages (from ultralytics) (2.3.5)
Requirement already satisfied: matplotlib>=3.3.0 in E:\OPG_env\opg_venv\Lib\site-packages (from ultralytics) (3.10.8)
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Requirement already satisfied: torchvision>=0.9.0 in E:\OPG_env\opg_venv\Lib\site-packages (from ultralytics) (0.21.0+cu124)
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  Downloading polars-1.44.1-py3-none-any.whl.metadata (11 kB)
Collecting nvidia-ml-py>=12.0.0 (from ultralytics)
  Downloading nvidia_ml_py-13.610.43-py3-none-any.whl.metadata (9.7 kB)
Collecting ultralytics-thop>=2.1.6 (from ultralytics)
  Downloading ultralytics_thop-2.1.6-py3-none-any.whl.metadata (13 kB)
Collecting ultralytics-platform>=0.1.3 (from ultralytics)
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Requirement already satisfied: charset_normalizer<4,>=2 in E:\OPG_env\opg_venv\Lib\site-packages (from requests>=2.23.0->ultralytics) (3.4.4)
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Requirement already satisfied: h11>=0.16 in E:\OPG_env\opg_venv\Lib\site-packages (from httpcore==1.*->httpx<1,>=0.28->ultralytics-platform>=0.1.3->
Requirement already satisfied: MarkupSafe>=2.0 in E:\OPG_env\opg_venv\Lib\site-packages (from jinja2->torch>=1.8.0->ultralytics) (2.1.5)
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Downloading ultralytics_platform-0.1.21-py3-none-any.whl (66 kB)
Downloading ultralytics_thop-2.1.6-py3-none-any.whl (30 kB)
Installing collected packages: nvidia-ml-py, polars-runtime-32, opencv-python, polars, ultralytics-thop, ultralytics-platform, ultralytics

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Successfully installed nvidia-ml-py-13.610.43 opencv-python-5.0.0.93 polars-1.44.1 polars-runtime-32-1.44.1 ultralytics-8.4.138 ultralytics-platform
```

In [4]:

```python
import cv2
import torch
from ultralytics import YOLO

# ------------------------------------------------
# Configuration
# ------------------------------------------------
from pathlib import Path

# ============================================================
# SETTINGS
# ============================================================

VIDEO_PATH = "E:\\OPG_env\\eye_work_28_aug_2026\\video_data\\video5.mp4"

OUTPUT_PATH = "E:\\OPG_env\\eye_work_28_aug_2026\\detected_video.mp4"


# YOLO11 models:
# yolo11n.pt = fastest
# yolo11s.pt = good balance
# yolo11m.pt = more accurate
# yolo11l.pt = higher accuracy
MODEL_PATH = "yolo11m.pt"

CONFIDENCE_THRESHOLD = 0.35
```

In [5]:

```python
# ------------------------------------------------
# GPU setup
# ------------------------------------------------

device = 0 if torch.cuda.is_available() else "cpu"

print("CUDA available:", torch.cuda.is_available())

if torch.cuda.is_available():
    print("GPU:", torch.cuda.get_device_name(0))
```

```
CUDA available: False
```

In [6]:

```python
# ------------------------------------------------
# Load YOLO model
# ------------------------------------------------

model = YOLO(MODEL_PATH)
```

```
[KDownloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11m.pt to 'yolo11m.pt': 100% ━━━━━━━━━━━━ 38.8MB 12.9MB/s 3
```

In [7]:
```
# ------------------------------------------------
# Open video
# ------------------------------------------------
cap = cv2.VideoCapture(VIDEO_PATH)

if not cap.isOpened():
    raise RuntimeError(f"Could not open video: {VIDEO_PATH}")

fps = cap.get(cv2.CAP_PROP_FPS)
width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
frame_count = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))

print(f"Resolution: {width}x{height}")
print(f"FPS: {fps}")
print(f"Frames: {frame_count}")
```

```
Resolution: 480x432
FPS: 30.0
Frames: 471
```

In [8]:
```
# ------------------------------------------------
# Output video writer
# ------------------------------------------------

fourcc = cv2.VideoWriter_fourcc(*"mp4v")

writer = cv2.VideoWriter(
    OUTPUT_PATH,
    fourcc,
    fps,
    (width, height)
)
```

In [ ]:
```
# ------------------------------------------------
# Object detection
# ------------------------------------------------

frame_number = 0

while True:

    ret, frame = cap.read()

    if not ret:
        break

    # Run YOLO inference

    results = model.predict(
    source=frame,
    device=device,
    conf=CONFIDENCE_THRESHOLD,
    imgsz=640,
    quantize=16 if torch.cuda.is_available() else 32,
    verbose=False
)

    result = results[0]

    # Draw bounding boxes, labels, confidence scores
    annotated_frame = result.plot()

    writer.write(annotated_frame)

    frame_number += 1

    if frame_number % 100 == 0:
        print(
            f"Processed {frame_number}/{frame_count} frames "
            f"({100 * frame_number / frame_count:.1f}%)"
        )
```

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In [ ]:

```
# ------------------------------------------------
# Cleanup
# ------------------------------------------------

cap.release()
```

```
writer.release()

print(f"Done. Saved to: {OUTPUT_PATH}")
```

In [ ]:

```
```

BASICS

Load YOLO and detect objects

Image

↓

YOLO model

↓

Object detection

↓

Bounding boxes + confidence + class

In [ ]:

```
from ultralytics import YOLO

model = YOLO("yolo11n.pt")

results = model(r"C:\Users\sufis\Downloads\img1.jpg")
#insert your own paths

for result in results:
    result.show()
```

```
[KDownloading https://github.com/ultralytics/assets/releases/download/v8.3.0/yolo11n.pt to 'yolo11n.pt': 100% ━━━━━━━━━━━━ 5.4MB 139.2KB/s 3

image 1/1 C:\Users\sufis\Downloads\img1.jpg: 448x640 1 banana, 2 apples, 4 oranges, 73.1ms
Speed: 8.5ms preprocess, 73.1ms inference, 73.6ms postprocess per image at shape (1, 3, 448, 640)
```

Look inside YOLO's result

In [ ]:

```
# Get the result for the first image

result = results[0]

print(result)
# print(result.boxes)
```

```
ultralytics.engine.results.Results object with attributes:

boxes: ultralytics.engine.results.Boxes object
keypoints: None
masks: None
names: {0: 'person', 1: 'bicycle', 2: 'car', 3: 'motorcycle', 4: 'airplane', 5: 'bus', 6: 'train', 7: 'truck', 8: 'boat', 9: 'traffic light', 10: 'f
obb: None
orig_img: array([[[255, 255, 255],
        [255, 255, 255],
        [255, 255, 255],
        ...,
        [255, 255, 255],
        [255, 255, 255],
        [255, 255, 255]],

       [[255, 255, 255],
        [255, 255, 255],
        [255, 255, 255],
        ...,
        [255, 255, 255],
        [255, 255, 255],
        [255, 255, 255]],

       [[255, 255, 255],
        [255, 255, 255],
        [255, 255, 255],
        ...,
        [255, 255, 255],
        [255, 255, 255],
        [255, 255, 255]],

       ...,

       [[255, 255, 255],
        [255, 255, 255],
        [255, 255, 255],
        ...,
        [255, 255, 255],
        [255, 255, 255],
        [255, 255, 255]],

       [[255, 255, 255],
        [255, 255, 255],
        [255, 255, 255],
        ...,
        [255, 255, 255],
```

```
        [255, 255, 255],
        [255, 255, 255]],

       [[255, 255, 255],
        [255, 255, 255],
        [255, 255, 255],
        ...,
        [255, 255, 255],
        [255, 255, 255],
        [255, 255, 255]]], shape=(408, 612, 3), dtype=uint8)
orig_shape: (408, 612)
path: 'image0.jpg'
probs: None
save_dir: 'C:\\Users\\sufis\\Downloads\\runs\\detect\\predict'
speed: {'preprocess': 4.460000083781779, 'inference': 51.57040001358837, 'postprocess': 6.317500025033951}
```

In [ ]:
```python
# Bounding-box coordinates
# Format:
# [x1, y1, x2, y2]
print(result.boxes.xyxy)
```

```
tensor([[357.6599, 215.8846, 506.3106, 362.4995],
        [275.6638,  42.3720, 546.9448, 250.2341],
        [141.1342, 244.1493, 277.2906, 379.3846],
        [294.0269, 121.0689, 435.6662, 251.2020],
        [ 90.8021,  84.5405, 301.3083, 248.4052],
        [ 96.1041, 197.6751, 229.2244, 344.6725],
        [230.2206, 198.6188, 373.0497, 296.7406]], device='cuda:0')
```

In [ ]:
```python
# Confidence score for each detected object

print(result.boxes.conf)
```

```
tensor([0.9062, 0.8777, 0.8586, 0.8079, 0.4101, 0.2771, 0.2502], device='cuda:0')
```

In [ ]:
```python
# Dictionary containing class IDs and class names

print(result.names)
```

```
{0: 'person', 1: 'bicycle', 2: 'car', 3: 'motorcycle', 4: 'airplane', 5: 'bus', 6: 'train', 7: 'truck', 8: 'boat', 9: 'traffic light', 10: 'fire hyd
```

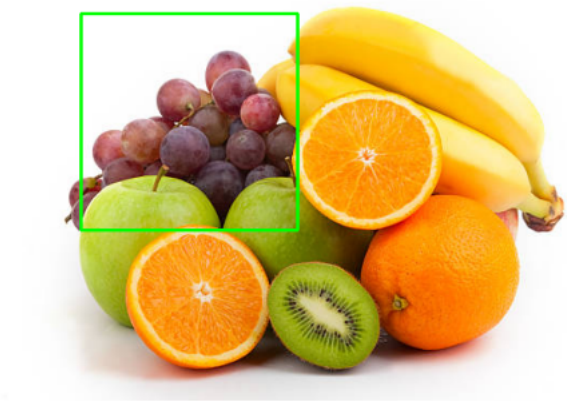
In [ ]:
```python
# Class ID of each detected object

print(results[0].boxes.cls)
```

```
tensor([49., 46., 49., 49., 47., 47., 49.], device='cuda:0')
```

In [ ]:
```python
# Display YOLO's detections

results[0].show()
```

Step 2 — Manually draw YOLO's bounding boxes

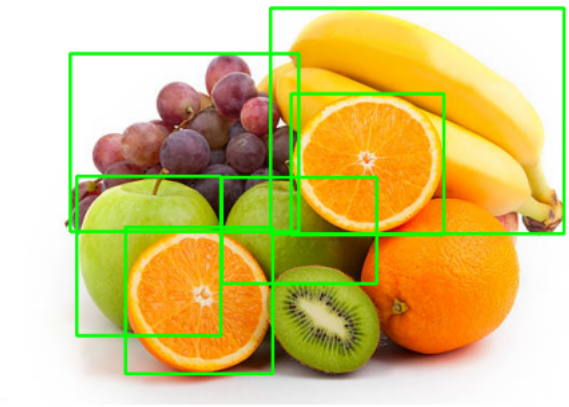
In [ ]:
```python
import cv2
import matplotlib.pyplot as plt

image = cv2.imread(r"C:\Users\sufis\Downloads\img1.jpg")

# Draw a rectangle manually
#
# (100, 50)  → top-left corner
# (300, 250) → bottom-right corner
# (0, 255, 0) → green color
# 2          → thickness

cv2.rectangle(
    image,
    (100, 50),
    (300, 250),
    (0, 255, 0),
    2
)

# OpenCV reads images in BGR format.
# Matplotlib expects RGB format.

image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.imshow(image)
plt.axis("off")
plt.show()
```



Replace manual coordinates with YOLO coordinates

In [20]:

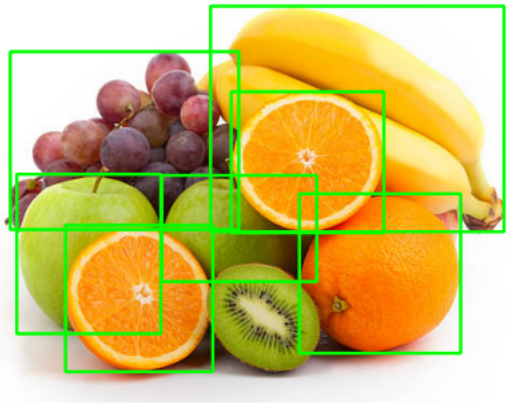
```
import cv2
import matplotlib.pyplot as plt

image = cv2.imread(r"C:\Users\sufis\Downloads\img1.jpg")

# Bounding boxes obtained from YOLO
boxes = [
    [275.6638, 42.3720, 546.9448, 250.2341],
    [141.1342, 244.1493, 277.2906, 379.3846],
    [294.0269, 121.0689, 435.6662, 251.2020],
    [90.8021, 84.5405, 301.3083, 248.4052],
    [96.1041, 197.6751, 229.2244, 344.6725],
    [230.2206, 198.6188, 373.0497, 296.7406]
]

# Draw a rectangle for every bounding box
for box in boxes:

    x1, y1, x2, y2 = box

    # Convert coordinates from float to integer
    x1 = int(x1)
    y1 = int(y1)
    x2 = int(x2)
    y2 = int(y2)

    cv2.rectangle(
        image,
        (x1, y1),
        (x2, y2),
        (0, 255, 0),
        2
    )

# Convert BGR to RGB for Matplotlib
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.imshow(image)
plt.axis("off")
plt.show()
```



In [21]:

```
from ultralytics import YOLO
import cv2
import matplotlib.pyplot as plt

# Load YOLO model
model = YOLO("yolo11n.pt")

# Read image
image = cv2.imread(r"C:\Users\sufis\Downloads\img1.jpg")

# Run YOLO on the image
results = model(image)

# Get the YOLO result
for result in results:

    # Get bounding boxes
    boxes = result.boxes.xyxy

    # Draw every bounding box
    for box in boxes:

        x1, y1, x2, y2 = box

        # Convert coordinates to integers
        x1 = int(x1)
        y1 = int(y1)
        x2 = int(x2)
        y2 = int(y2)

        cv2.rectangle(
            image,
            (x1, y1),
            (x2, y2),
            (0, 255, 0),
            2
        )

# Convert BGR to RGB
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Display image
plt.imshow(image)
plt.axis("off")
plt.show()
```

```
0: 448x640 1 banana, 2 apples, 4 oranges, 51.7ms
Speed: 4.1ms preprocess, 51.7ms inference, 6.0ms postprocess per image at shape (1, 3, 448, 640)
```



YOLO finds where the objects are. OpenCV draws the boxes around those objects.

In [ ]: